

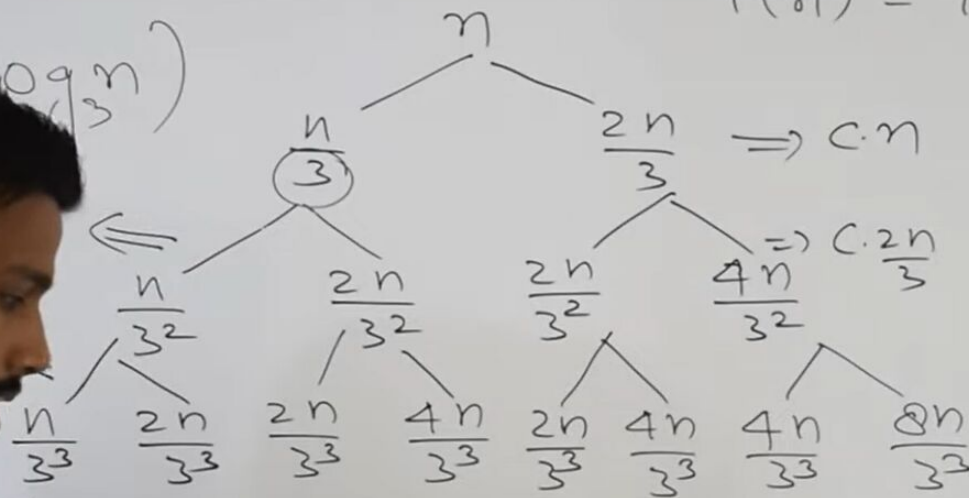
$$T(n) = T(n/3) + T(2n/3) + cn$$

$$T(2n/3) = T(2n/3^2) + T(4n/3^2) + c \cdot 2n/3$$

$$\frac{cn}{3} + \frac{c \cdot 2n}{3} \Rightarrow cn$$

$$\frac{n}{3^2} + \frac{2n}{3^2} + \frac{2n}{3^2} + \frac{4n}{3^2} = \frac{9n}{3^2} = cn$$

$(\log_{3/2} n)$



$$\left(\frac{n}{3^k}\right) = 1$$

$$n \Rightarrow \frac{n}{(3/2)^k} \Rightarrow \frac{n}{(3/2)^2} \Rightarrow \frac{n}{(3/2)^3} \dots \leq cn(1)$$

$$cn \cdot \log_{3/2} n = O(n \log_{3/2} n)$$

$$\left(\frac{n}{(3/2)^k}\right) = 1$$

$$k = \log_{3/2} n$$